

Data Sourcing & API Market Research

Individual Study Plan (IPS) / Data Aggregator Application

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Scope: Environmental & Weather Data · Geographic & Geocoding · Astronomy & Solar/Lunar Data

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Preface

This document provides a comprehensive survey of the commercial and open-data API landscape across three data categories required by the Context-Aware Data Aggregator project. The analysis covers providers originating from both the European Union (and EU-equivalent jurisdictions such as Switzerland and the UK) and the United States, addressing the practical differences in regulatory exposure, data residency, pricing structures, and suitability for a mobile application context.

Pricing figures were sourced from official provider websites in June 2026. Because API pricing — particularly for mid-tier SaaS providers — changes frequently, all figures should be treated as directionally accurate and verified against the provider's current pricing page before committing to any integration.

Each section concludes with a justified recommendation for use within the scope of this project: a student-grade mobile application with low-to-moderate API call volumes, no dedicated infrastructure budget, and a need for clear commercial licensing even at the free tier.

Category 1: Environmental & Weather Data

Market Landscape

The weather API market broadly splits into three tiers. At the top sit enterprise-grade platforms like AccuWeather and Tomorrow.io, which derive competitive advantage from proprietary forecast models, global sensor networks, and institutional data agreements. In the middle sits a crowded band of mid-market providers — OpenWeatherMap, Weatherbit, Visual Crossing — that bundle reasonable forecast quality with self-service pricing. At the bottom (in terms of pricing, not necessarily quality) are the open-data newcomers, most notably Open-Meteo, which has disrupted the space by offering a genuinely free tier for non-commercial use backed by WMO-member national weather service data.

From a regulatory standpoint, the split between EU and US providers is meaningful here. Most leading commercial providers are US-based, which creates a GDPR consideration if the application processes user data alongside API calls (e.g., sending a user's GPS coordinates to a weather endpoint). Weather APIs themselves typically only receive coordinates and return atmospheric data — no personal data is processed by the API in most cases — but the terms and jurisdiction of the provider still affect how you document third-party data flows in a privacy notice.

Provider Profiles

Open-Meteo | *Switzerland*

Open-Meteo is a Swiss open-source project that aggregates forecast data from national weather services including DWD (Germany), NOAA (US), ECMWF (EU), and others. The free tier is genuinely generous — 10,000 API calls per day and 300,000 per month — but it is explicitly restricted to non-commercial use. For commercial use, the company offers a subscription model starting at **\$29/month** for 1 million calls (Standard) and **\$99/month** for 5 million calls (Professional). Enterprise contracts (50M+ calls/month) are available on request. All paid tiers run on dedicated infrastructure with a **99.9% uptime target**. No credit card is required for the free tier.

Switzerland is considered a GDPR-adequate country under Article 45, meaning data transfers are treated similarly to intra-EU transfers. Open-Meteo does not process personal data — it only receives coordinates — so the GDPR exposure here is minimal regardless.

Project suitability: Excellent. The free tier is sufficient for development and low-volume production. If the project is classified as non-commercial (which a university submission typically is), no payment is required at all. The upgrade path to paid is clean and affordable.

OpenWeatherMap (OpenWeather Ltd.) | *United Kingdom*

OpenWeatherMap is arguably the most widely integrated weather API in the developer ecosystem, which means it has extensive community documentation, SDK support, and troubleshooting resources. The base free tier offers **1,000 API calls per day** and **1,000,000 per month**, covering current weather, a 5-day/3-hourly forecast, air pollution data, and geocoding. This tier has a **95% uptime SLA** and data updates every 2 hours.

The more interesting product for mobile apps is the **One Call API 3.0**, which returns a comprehensive weather package — current conditions, hourly, daily, and minutely forecasts — in a single request. The first **1,000 calls/day** are free, but a credit card is required on file, and anything above that is billed at **€0.14 per 100 calls**. Paid subscription plans begin at approximately **\$40/month** (Startup tier, 100k calls/month), stepping up to \$140/month, \$470/month, and \$2,000/month for higher volumes.

The UK received an adequacy decision from the EU after Brexit, meaning GDPR-equivalent protection is recognized. OpenWeather provides a Data Processing Agreement (DPA) for business users.

Project suitability: Good. The free tier is usable and well-documented. The One Call API 3.0 is attractive because it reduces the number of requests needed per user session, though the credit card requirement for any overage is a friction point for a student project.

AccuWeather | *United States*

AccuWeather is an institutional-grade provider best known for its MinuteCast precipitation-by-the-minute product and its RealFeel temperature index, both of which are proprietary. Unfortunately, neither the pricing nor the free tier is developer-friendly. There is no permanent free tier — the 500 calls/day allowance is part of a **14-day trial only**. After the trial, the minimum paid plan is **\$2/month** (Starter, 15,000 calls/month), but a notable catch applies: *every weather lookup costs two API calls* because location resolution and weather retrieval are billed separately. This effectively halves any stated limit in practice.

The Standard plan is **\$25/month** (225,000 monthly calls), Prime is **\$250/month** (1.8M calls), and Elite is **\$500/month** (2.4M calls). MinuteCast is billed as a separate product on top of these. SLA terms for self-service plans are not publicly documented; enterprise contracts offer custom guarantees. As a US company, transfers to AccuWeather would require Standard Contractual Clauses (SCCs) under GDPR.

Project suitability: Poor for this project. The absence of a permanent free tier, the double-billing model, and US jurisdiction with no published SLA make this a poor fit for a low-budget educational application.

Tomorrow.io | *United States*

Tomorrow.io has built its reputation on hyperlocal forecasting driven by proprietary ML models and satellite-derived data layers. The platform is genuinely impressive technically — it offers 60+ data layers including pollen, air quality, fire weather index, and lightning in addition to standard meteorological parameters. The free tier allows **500 API calls per day**, covering a 5-day forecast with core weather parameters for a single monitored location.

The problem is pricing opacity. Tomorrow.io's paid API plans are quote-based and require contacting sales; public sources suggest pricing starts in the range of **\$500/month**, placing it firmly in the commercial/enterprise bracket. The homepage claims **99.9% uptime**, and enterprise contracts include custom SLAs. No EU data residency is mentioned. As a US entity, GDPR compliance relies on SCCs.

Project suitability: Marginal. The free tier (500 calls/day) is sufficient for a university project's development and demonstration phases. The data quality is excellent. However, any move to a paid tier for this project is not practical, and the commercial licensing terms on the free tier should be verified if the project is framed as commercially oriented.

Weatherbit | *United States*

Weatherbit offers a serviceable mid-tier weather API with a particular strength in historical data access — paid plans unlock up to 25 years of historical weather records. The free tier is tight: **50 calls/day**, restricted to non-commercial use only. A 21-day free trial of paid features is available without a credit card, which is a reasonable way to evaluate the platform. Paid plans begin at approximately **\$35/month** for the Standard tier (25,000 calls/day, 16-day forecast, commercial license), stepping up through Plus and Business tiers. Uptime SLA ranges from 95% (free) to 99.5%+ (Business). US jurisdiction; SCCs required for GDPR.

Project suitability: Poor for this project. The 50 calls/day free tier is too restrictive for any meaningful testing, and the non-commercial restriction on the free tier is an additional constraint.

Visual Crossing | *United States*

Visual Crossing is a strong contender that often gets overlooked in favour of higher-profile brands. Its defining advantage is a free tier that is both generous and commercially licensed: **1,000 records/day at no cost**, no credit card required, and commercial use is explicitly permitted. One "record" equates to one day of weather data for one location, which is efficient for daily-granularity use cases. Pay-as-you-go pricing above the free tier is **\$0.0001 per record** — remarkably cheap. A Professional Plan (single-user, unlimited records) starts at approximately **\$35/month**, and corporate/enterprise plans are available for high-volume use.

The platform includes historical weather going back decades, which is accessible even on the free tier, and this is genuinely rare at this price point. It is a US company with no EU data residency, though GDPR exposure is low given that only coordinates are shared with the API.

Project suitability: Good. The commercially-licensed free tier is ideal for a university project that may be presented in a commercial framing. Historical data access is a bonus if the app involves any retrospective analysis.

Meteomatics | *Switzerland*

Meteomatics is a premium meteorological services company based in Liechtenstein/Switzerland, serving aviation, energy, and insurance sectors. Its weather API has exceptional spatial and

temporal resolution, drawing from proprietary observation networks and numerical weather prediction models. A 14-day trial is available. Paid pricing is enterprise-only and entirely quote-based — there are no publicly listed tiers. Typical contracts are reported to be in the thousands of euros per month range. Switzerland is a GDPR-adequate country.

Project suitability: Not applicable for this project. The cost and enterprise sales process make Meteomatics unsuitable for a student application, though it is worth acknowledging as the premium end of the EU-adjacent market.

NOAA Weather API (api.weather.gov) | *United States Government*

The National Oceanic and Atmospheric Administration offers a completely free REST API with no key requirement and no formal rate limits (though respectful usage is expected). The data quality is authoritative — it is the primary source for official US weather forecasts — and there is no commercial restriction.

Project suitability: Not applicable. NOAA’s API covers the United States exclusively. If the application is expected to function in Europe or globally, NOAA is not a viable primary data source. It is mentioned here for completeness as the dominant free offering in the US market.

Copernicus Climate Data Store / ECMWF | *European Union*

The Copernicus Climate Data Store (CDS), operated by the ECMWF on behalf of the EU, provides free access to a vast archive of climate reanalysis data, seasonal forecasts, and satellite-derived datasets. Access requires free registration (name, email, country, sector) and agreement to per-dataset terms of use. The data is global in coverage and authoritative in quality.

Project suitability: Not applicable as a primary weather API. CDS data access is batch-oriented via a Python API client or web interface — there is no REST endpoint that returns “current conditions for coordinates.” It is an important academic resource for climate data analysis but does not fit the real-time mobile application use case.

Weather API Comparison Table

Provider	Origin	Free Tier	Paid Entry Tier	Commercial on Free?	SLA (Paid)	GDPR Posture
Open-Meteo	Switzerland	10,000/day 300,000/mo	\$29/mo (1M calls)	Non-commercial only	99.9%	Adequate country
OpenWeatherMap	UK	1,000/day 1M/mo	~\$40/mo (100k calls)	Yes	99.5%	UK adequacy decision
AccuWeather	US	500/day (trial only, 14 days)	\$2/mo (15k/mo)	No permanent free tier	Not published	SCCs required
Tomorrow.io	US	500/day	~\$500/mo (quote-based)	Yes (verify TOS)	99.9% (claimed)	SCCs required
Weatherbit	US	50/day	~\$35/mo (25k/day)	Non-commercial only	99.5%	SCCs required

Provider	Origin	Free Tier	Paid Tier	Entry Commercial on Free?	SLA (Paid)	GDPR Posture
Visual Crossing	US	1,000 records/day	~\$35/mo (unlimited records)	Yes	Not published	SCCs required
Meteomatics	Switzerland	14-day trial	Quote-based (enterprise)	N/A	Custom	Adequate country
NOAA	US Gov	Unlimited (US coverage only)	Free	Public domain	None	N/A
Copernicus / ECMWF	EU	Free (registration required)	Free	Open license	None	EU-native

Weather Data Recommendation

Primary: Open-Meteo (free tier for non-commercial / Standard \$29/month for commercial)

Open-Meteo is the correct choice for this project. It aggregates data from multiple national weather services (ECMWF, DWD, NOAA) into a single unified REST API, delivering forecast quality that competes with paid providers. The free tier of 10,000 calls/day is more than sufficient for any realistic student project load. If the project is submitted or deployed in a commercial context, the upgrade path to the Standard plan at \$29/month is the most cost-effective commercial license in this market. The Swiss jurisdiction is a clean fit for EU projects. No credit card is required for any free-tier use.

Alternative: Visual Crossing

If commercial licensing on the free tier is a hard requirement from day one, Visual Crossing's 1,000 records/day free commercial tier is the strongest fallback. It is less convenient for high-frequency real-time lookups (1 record = 1 day of data), but it is well-suited to any use case involving forecast data or historical weather.

Category 2: Geographic & Locality Context / Geocoding

Market Landscape

Geocoding — converting between coordinates and human-readable location descriptions — is a market with enormous variation in pricing, licensing, and data quality. The key distinction between providers is not just cost but *what you can do with the results*: some providers prohibit storing geocoded results, which is a meaningful constraint for a mobile app that might want to cache location names for offline use. The data source matters too — most providers either license data from HERE or TomTom's proprietary datasets, or derive results from OpenStreetMap's openly licensed map data. The latter generally permits far more permissive downstream use.

From a GDPR perspective, geocoding APIs are one of the most sensitive categories: every API call includes the user's physical coordinates, which constitute location data and are classified as

personal data under GDPR. This means the jurisdiction and data processing terms of the geocoding provider are directly relevant, not just incidental.

Provider Profiles

Google Maps Platform — Geocoding API | *United States*

Google is the market leader in geocoding coverage and result accuracy, particularly for addresses in emerging markets and rural areas where OSM coverage is thin. In March 2025, Google restructured its billing model significantly, replacing the previous \$200/month credit with per-SKU free usage thresholds. The Geocoding API now includes **10,000 free requests/month**, after which pricing is **\$5 per 1,000 requests**. Optional subscription plans are available: Starter (\$100/month, 50k combined calls), Essentials (\$275/month, 100k), and Pro (\$1,200/month, 250k). The API enforces a rate limit of 3,000 requests/minute. A credit card is required to create an account. The uptime SLA is **99.9%**, backed by Google's infrastructure commitments.

Google provides a DPA and Standard Contractual Clauses for EU users, meaning GDPR compliance is achievable but requires configuration of your Google Cloud account's data processing settings.

Project suitability: Acceptable but not ideal. The free tier is usable, but the mandatory credit card and the US jurisdiction make this more friction than necessary for a university project. The pricing is also the highest in this category at scale.

Mapbox | *United States*

Mapbox occupies the middle ground between Google's comprehensive commercial offering and the open-data providers. Its geocoding free tier is generous — **100,000 requests/month** — and no credit card is required. Beyond that, temporary geocoding (no result storage) costs **\$0.75 per 1,000 requests**, dropping to \$0.45/1000 at volumes above 1 million. Permanent geocoding (where results may be stored) costs \$5/1000 after the free tier, which is on par with Google.

Mapbox also uses a mix of OpenStreetMap and proprietary data, and its licensing reflects this complexity — the storage restriction is the key one to watch for a mobile app. No formal uptime SLA is published for non-enterprise plans. US jurisdiction; DPA available.

Project suitability: Good for the free tier. 100,000 requests/month is very comfortable for a student project, and the no-credit-card policy makes signup straightforward.

HERE Technologies | *Germany (European Union)*

HERE is a German company — formed from the former Nokia Maps business — and is the most significant EU-origin player in the commercial geocoding market. It covers 200+ countries and provides geocoding, reverse geocoding, search, and routing. The Freemium plan offers a very generous **250,000 free requests/month** (note: some plan configurations show 30,000/month for certain account types — verify at signup). After the free tier, pricing is **\$1 per 1,000 requests**, and the Pro plan (\$449/month) includes 1 million transactions with priority support. Rate limits on the Freemium plan are approximately 5 requests/second.

HERE has published an explicit GDPR compliance statement and has the required DPAs in place. As a German company, it operates under EU law natively, which substantially simplifies data processing documentation for an EU-based project.

Project suitability: Excellent. The combination of a generous free tier, EU legal jurisdiction, and competitive paid pricing makes HERE the strongest geocoding option for a European project. 250,000 free requests/month is far more than this project would ever consume.

BigDataCloud | *Australia*

BigDataCloud occupies a unique niche: it offers a **client-side reverse geocoding API that is permanently free, with no documented rate limit**, on the condition that it is called directly from the user's device for the device's current location obtained with that user's consent. This is not just a "technically free" tier — it is a deliberate product decision; the company explains that client-side calls carry the user's IP and coordinates together, which allows BigDataCloud to monetise the anonymised aggregate data rather than charging the developer.

For server-side reverse geocoding (coordinates from stored data or third-party sources), the free tier is 10,000 requests/month, with paid plans from \$29.95/month (100k) up to \$1,999/month (10M). BigDataCloud is an Australian company; Australia has an EU adequacy decision under some frameworks, though this is not as straightforward as an EU-native provider.

Project suitability: Very good for the reverse geocoding use case specifically. The client-side free API is almost perfectly matched to a mobile app that needs to resolve the user's current coordinates into a place name. The constraint (must be called client-side, for the user's own location, with consent) aligns naturally with how a context-aware app would actually use reverse geocoding.

Nominatim / OpenStreetMap Foundation | *United Kingdom (OSMF)*

Nominatim is the geocoding engine behind OpenStreetMap. It is open-source, free, and uses OSM data — meaning results can be cached and reused without licensing concern. However, the OSM Foundation's publicly hosted Nominatim instance (nominatim.openstreetmap.org) enforces a strict usage policy: **maximum 1 request per second**, no bulk queries, and critically — *applications whose primary function involves geocoding must run their own instance*. This is a common point of misunderstanding; many developers assume Nominatim is freely available for production use, but the hosted instance is a community courtesy, not a production service.

For this project, the public Nominatim instance is usable during development, but deploying a mobile app against it would be a policy violation if geocoding is a primary feature.

Project suitability: Limited. Acceptable for development and low-volume personal use. Not suitable for production deployment without self-hosting. The OSM Foundation strongly recommends commercial Nominatim hosting providers for production applications.

Geocode.Earth (hosted Pelias) | *United States*

Pelias is an open-source geocoder built on Elasticsearch, maintained by the Geocode.Earth team. Geocode.Earth is the hosted, managed version of Pelias, using fully open data (OpenStreetMap, OpenAddresses, Who's on First, Geonames). Because the underlying data has permissive licensing, geocoded results can be stored without restriction — a significant advantage over Google and Mapbox. The free tier provides **25,000 requests/month** with no credit card required. Paid plans start at **\$9.99/month** (100k requests), rising to \$29.99/month (500k) and \$99.99/month (2M). US jurisdiction.

The self-hosting path (running Pelias on your own infrastructure) is entirely free and places no data residency constraints at all, making it arguably the most GDPR-friendly option if you have access to an EU cloud server.

Project suitability: Good. The free tier is comfortable and the no-storage-restriction licensing is a real advantage. The \$9.99/month paid tier is accessible even on a student budget.

Radar.io | *United States*

Radar is a geofencing and location platform that includes geocoding as one of its features. Its geocoding free tier is generous — **100,000 requests/month** — with paid pricing at **\$0.50 per 1,000 requests**, making it the most cost-competitive among the major US providers. US jurisdiction; DPA available. No formal SLA published for non-enterprise tiers.

Project suitability: Good. Competitive pricing and a generous free tier. More suitable if the application uses geofencing or location tracking alongside geocoding.

Azure Maps | *United States (Microsoft)*

Azure Maps is Microsoft's cloud geocoding service, built on TomTom data. The free tier is limited — **5,000 requests/month** — with pay-as-you-go pricing at **\$4.50 per 1,000 requests** (up to 500k). The meaningful differentiator for EU projects is that Azure Maps can be configured to process data in EU Azure regions, which provides data residency guarantees without routing coordinates to US data centres. Microsoft provides a comprehensive DPA as part of its Online Services Terms. Note that the Gen1 pricing tier is being retired in September 2026; all accounts should be on Gen2.

Project suitability: Poor for this project due to the small free tier (5,000 requests/month). Better suited to existing Azure-ecosystem projects where the data residency configuration has value.

TomTom | *Netherlands (European Union)*

TomTom is a Dutch company with deep roots in navigation data — it is the underlying data provider for a number of other APIs on this list, including Azure Maps. Its developer API offers a Freemium plan with a combined daily allowance of **2,500 non-tile requests/day** (approximately 75,000/month) across all products. Paid pricing is **€0.50 per 1,000 requests**. Crucially, commercial use is explicitly permitted even on the Freemium plan, and no credit card is required. TomTom is a publicly listed Dutch company operating fully under EU law, with GDPR compliance baked in by default.

Project suitability: Strong. TomTom provides an EU-native geocoding option with commercial-friendly free tier terms. The daily limit (2,500 requests) is generous for a student project. For a project with a European focus, TomTom and HERE are the two best choices.

Geocoding API Comparison Table

Provider	Origin	Free Tier	Paid Entry	Storage Rights	CC Required	GDPR Posture
Google Maps	US	10,000/mo	\$5/1,000	Restricted	Yes	SCCs (DPA available)
Mapbox	US	100,000/m	\$0.75/1,00	Conditional	No	SCCs (DPA

Provider	Origin	Free Tier	Paid Entry	Storage Rights	CC Required	GDPR Posture
		0	0 (temp) \$5/1,000 (stored)			available)
HERE Technologies	Germany	250,000/mo	\$1/1,000	Yes	No	EU-native, GDPR statement published
BigDataCloud	Australia	Client-side: unlimited free Server-side: 10k/mo	\$29.95/mo (100k)	Yes	No	Australian adequacy (verify)
Nominatim / OSM	UK (OSMF)	1 req/sec (hosted)	Self-host (free)	OSM license	No	EU-adequate (public instance, UK)
Geocode.Earth	US	25,000/mo	\$9.99/mo (100k)	Yes (open data)	No	SCCs; self-host option for EU
Radar.io	US	100,000/mo	\$0.50/1,000	Yes	No	SCCs (DPA available)
Azure Maps	US (Microsoft)	5,000/mo	\$4.50/1,000	Yes	Yes	EU data residency available
TomTom	Netherlands	2,500 req/day (~75k/mo)	€0.50/1,000	Yes	No	EU-native, GDPR compliant

Geocoding Recommendation

Primary reverse geocoding: BigDataCloud (client-side free tier)

For the core use case of this application — “what location context is the user currently in?” — BigDataCloud’s client-side reverse geocoding API is an almost perfect fit. The API is called from the device, resolves the user’s current coordinates into a structured locality (city, region, country, administrative subdivision), and does so with no rate limit and no cost. The client-side constraint aligns naturally with the architecture of a mobile app.

Secondary / forward geocoding: HERE Technologies or TomTom

If the application requires any forward geocoding (converting a place name to coordinates) or server-side reverse geocoding, HERE Technologies is the recommended choice: the free tier of 250,000 requests/month is exceptionally generous, the pricing per-call is competitive, results can be stored, and it is an EU-native company with native GDPR compliance. TomTom is a credible alternative if EU jurisdiction is the top priority and the slightly lower free tier volume (75k/month) is sufficient.

Category 3: Astronomy & Solar/Lunar Data

Market Landscape

The astronomy data API market is distinctly different from the previous two categories: it is far smaller commercially, dominated by either government/academic sources (free, authoritative, no SLA) or lightweight independently operated services (free, no key, attribution-only). Dedicated commercial astronomy APIs are few; most providers enter this space as an add-on feature to a geolocation or weather product rather than as a standalone offering.

For a mobile context-aware application, the most practically relevant data is: sunrise and sunset times, civil/nautical/astronomical twilight boundaries, moon phase, and solar noon. Anything beyond that (planetary ephemerides, eclipse predictions, deep-sky object positions) is academic territory and unlikely to serve user-facing features in this type of application.

Provider Profiles

SunriseSunset.io | *Independent / Unknown Jurisdiction*

SunriseSunset.io is a lightweight, independently operated API delivering sunrise, sunset, solar noon, golden hour, civil/nautical/astronomical twilight, day length, and moon phase data for any coordinate pair. No API key is required. No registration is required. Commercial use is permitted with attribution — specifically, a link back to SunriseSunset.io must appear on any page or interface that displays the data. There are no published rate limits, no SLA, and no pricing whatsoever. The API is called with a latitude, longitude, and optional date parameter.

The operator's jurisdiction is not clearly documented, which is the main reservation. The service has no uptime commitment and has historically experienced occasional brief outages.

Project suitability: Well-suited for this project during development and demonstration. The attribution requirement is easily satisfied in a mobile UI footer or credits screen. For production deployment, the lack of SLA is a risk that should be acknowledged.

Sunrise-Sunset.org | *Independent / Unknown Jurisdiction*

Very similar to SunriseSunset.io in scope and terms. Free, no API key, attribution required. The data set is slightly more limited — sunrise, sunset, and twilight times — without the golden hour or moon phase fields. The two services are broadly interchangeable for basic use.

Project suitability: Good as a fallback to SunriseSunset.io, or for use in parallel to verify results.

ipgeolocation.io — Astronomy API | *Global / US-registered entity*

ipgeolocation.io is a commercial geolocation platform that offers astronomy data as part of its API suite. The Astronomy API is the most feature-complete paid-tier option in this category, providing: sunrise, sunset, solar noon, moonrise, moonset, moon phase, sun and moon altitude and azimuth, distance from Earth, illumination percentage, parallactic angle, civil/nautical/astronomical twilight, blue hour, and golden hour.

The free tier offers **1,000 API credits per day** (shared across all ipgeolocation.io APIs). Paid plans begin at **\$19/month** for 150,000 credits/month, with Plus at \$49/month (500k), Pro at \$79/month (1M), and Business at \$129/month (2M). Annual billing provides approximately 17%

savings. The Astronomy API can also be queried with an explicit date to retrieve historical or future solar/lunar data.

The company's GDPR compliance page exists but contains limited detail on data residency locations.

Project suitability: Very good as a commercial option. The free tier (1,000 credits/day) is ample for this project, and the breadth of data returned per call is the highest in this category. If the project requires detailed moon data or golden hour/blue hour times, this is the strongest option among free-tier providers.

USNO Astronomical Applications API (US Naval Observatory) | *United States Government*

The US Naval Observatory's Astronomical Applications Department provides a free REST API covering sunrise, sunset, moonrise, moonset, civil/nautical/astronomical twilight, moon phases, Julian date conversion, and eclipse predictions. No API key is required, though large-volume users are encouraged to register their application identifier via email. The data is authoritative — the USNO is the official source of astronomical data for US government purposes — and global in coverage.

There are no formal rate limits, though USNO documentation recommends courtesy usage. As a product of the US federal government, the data is in the public domain and there are no restrictions on commercial use.

The meaningful caveats are: no uptime SLA (it is a government service, not a commercial product), and its status as a US Department of Defense asset raises a faint question about long-term availability and access from non-US contexts, though in practice it has been reliably accessible globally.

Project suitability: Good as a secondary or verification source. The authoritative quality and public domain license are strong points. The lack of SLA and DoD affiliation make it a poor primary source for a production application.

JPL Horizons System (NASA/JPL) | *United States Government*

NASA's Jet Propulsion Laboratory Horizons system provides precise ephemerides for solar system bodies — positions, velocities, and physical parameters for planets, moons, asteroids, and comets. It is entirely free and in the public domain. The API interface, however, is designed for scientific batch queries rather than real-time application calls.

Project suitability: Not applicable. JPL Horizons is research-grade infrastructure. There is no use case in a context-aware mobile application that would justify this level of astronomical precision, and the API interface is not designed for real-time mobile integration.

Heavens-Above | *Germany (Individual developer)*

Heavens-Above is a German website operated by software developer Chris Peat, providing satellite pass predictions, ISS visibility, and general astronomical data. It has no documented public API — the site provides a web interface only.

Project suitability: Not applicable. No programmable API is available.

Visual Crossing — Astronomy Component | *United States*

Visual Crossing's weather API includes solar and lunar data fields (sunrise, sunset, moonrise, moonset, moon phase) as part of the standard weather response object. This means if you are already using Visual Crossing for weather, astronomy data arrives at no additional cost within the same API call. It does not provide the granular twilight timing, blue/golden hour, or solar position data that a dedicated astronomy API offers.

Project suitability: Convenient if Visual Crossing is already in the stack. Not sufficient as a standalone astronomy data source.

Astronomy API Comparison Table

Provider	Origin	Free Tier	Paid Entry	Key Required	Commercial Use	Data Coverage	SLA
SunriseSunset.io	Unknown	Unlimited (attribution)	Free	No	With attribution	Rise/set, twilight, golden hour, moon phase	None
Sunrise-Sunset.org	Unknown	Unlimited (attribution)	Free	No	With attribution	Rise/set, twilight	None
ipgeolocation.io	Global/US	1,000 credits/day	\$19/mo (150k/mo)	Yes	Yes	Most comprehensive — twilight, position, moon detail	Not published
USNO	US Gov	Unlimited	Free (public domain)	No	Public domain	Rise/set, twilight, phases, eclipses	None (gov service)
JPL Horizons	US Gov	Unlimited	Free (public domain)	No	Public domain	Full solar system ephemerides	None
Heavens-Above	Germany	N/A	N/A	N/A	N/A	Satellite passes, sky events	N/A — no API
Visual Crossing	US	1,000 records/day	~\$35/mo (unlimited)	Yes	Yes	Basic rise/set + moon phase only	Not published

Astronomy Recommendation

Primary: SunriseSunset.io

For the scope of a context-aware mobile application, SunriseSunset.io provides everything needed: sunrise, sunset, golden hour, civil twilight, and moon phase in a single zero-cost, zero-friction call. The attribution requirement is the only constraint and is easy to satisfy. It is the recommendation for this project.

Upgrade path: ipgeolocation.io Astronomy API

If the application requires richer solar/lunar context — moon illumination percentage, solar azimuth for AR features, blue hour timing for photography context — ipgeolocation.io's 1,000 free daily credits are sufficient for production use at student scale, and the \$19/month entry tier is accessible if commercialisation is ever pursued. This API also provides a single call that delivers all data fields at once, which is efficient for mobile data budgets.

Cross-Cutting Concerns: GDPR, Jurisdiction, and Data Residency

The EU/US Divide in Practice

For a mobile application processing user location, the EU vs. US origin of an API provider is not merely academic. Under GDPR, any transfer of personal data (including GPS coordinates) to a third-country entity — which includes US companies — requires a legal transfer mechanism. In practice, this means either:

1. **An adequacy decision** — the EU Commission has decided the third country provides equivalent protection (Switzerland , UK , Australia — partial/complex, US — no general adequacy decision).
2. **Standard Contractual Clauses (SCCs)** — contractual safeguards that must be in place between the developer and the US-based provider. Most major US API providers (Google, Mapbox, Tomorrow.io) provide SCCs as part of their DPA.
3. **Using EU-native providers** — eliminates the cross-border transfer issue entirely.

For a university student project, the practical implication is: - Using an EU-native provider (HERE, TomTom, Open-Meteo) is the cleanest path. - Using a US provider with an available DPA/SCC (Google, Mapbox, OpenWeatherMap) is legally compliant but requires including a data processing section in the app's privacy notice. - Using a US provider with no published DPA (smaller independent services, some free-tier providers) creates a documentation gap that is acceptable for a non-deployed academic project but would be a blocker for any real commercial deployment.

Summary of Jurisdictions

Category	EU/Adequate	US (SCC needed)	Unknown/Other
Weather	Open-Meteo (CH), Copernicus (EU)	AccuWeather, Tomorrow.io, Weatherbit	—
Weather adequacy)	(UK OpenWeatherMap	Visual Crossing	—
Geocoding	HERE (DE), TomTom (NL)	Google, Radar.io Mapbox,	BigDataCloud (AU)
Astronomy	—	USNO, ipgeolocation.io	JPL, SunriseSunset.io, Sunrise-Sunset.org

Recommended API Stack for This Project

Based on the analysis above, the following stack balances data quality, developer experience, cost, and GDPR exposure optimally for a university-level mobile application:

Data Category	Recommended Provider	Rationale
Weather — current & forecast	Open-Meteo (free tier)	10,000 calls/day, no credit card, EU-adjacent jurisdiction, non-commercial use free, excellent data quality from WMO-member model ensembles
Weather fallback	Visual Crossing (free commercial tier)	1,000 records/day with commercial license, zero friction signup, PAYG upgrade path
Reverse (primary)	Geocoding BigDataCloud (client-side free)	Unlimited free for device-location lookups, perfectly matched to mobile use case, no key required
Geocoding / (fallback)	server-side HERE Technologies (Freemium)	250,000 requests/month free, EU-native jurisdiction, result storage permitted
Astronomy / Solar data	SunriseSunset.io (free + attribution)	Zero setup, commercial OK, covers all primary solar context fields for a mobile app
Astronomy (enhanced, if needed)	ipgeolocation.io (1,000 credits/day free)	Moon detail, solar position, golden/blue hour — most complete free astronomy endpoint

This stack requires zero monthly spend for a typical student project with moderate usage, produces no hard data residency conflicts (Open-Meteo and BigDataCloud's client-side API both handle this cleanly), and offers a documented upgrade path for each component if the project is developed commercially post-graduation.